```
> # Importing Data -----------------------
> MoosePopulation <- read.csv("C:/Users/KatieTurpin/Downloads/MoosePopulation.csv")
> SaplingStudy <- read.csv("C:/Users/KatieTurpin/Downloads/SaplingStudy.csv")
> #
> # Title: Bio 1002 R assignment -------
> # Author: Katie Turpin
> # Date: 2026-02-13
> #
> # Set working directory ---------------
> setwd("/Users/KatieTurpin/Documents/R_Bio_1002")
> #
> # QUESTION 1 ==========================
> # Installing dplyr --------------------
> install.packages("dplyr")
> #
> # Loading dplyr -----------------------
> library(dplyr)
> #
> # QUESTION 2 ==========================
> # Naming data -------------------------
> moosedata <- read.csv("C:/Users/KatieTurpin/Downloads/MoosePopulation.csv")
> #
> # QUESTION 3 ==========================
> # Analyzing and omitting data ---------
> View(moosedata)
> na.omit(moosedata)
                 Ecoregion Year  Area
1        North_Shore_Forests 2020  8400
2  Northern_Peninsula_Forests 2020 12000
3         Long_Range_Barrens 2020  8800
4            Central_Forests 2020 20500
5            Western_Forests 2020  6800
6  EasternHyperOceanicBarrens 2020  4400
7           Maritime_Barrens 2020 21300
8             Avalon_Forests 2020  4200
9   StraitOfBelleIsleBarrens 2020   500
10       North_Shore_Forests 1992  8400
11 Northern_Peninsula_Forests 1992 12000
12        Long_Range_Barrens 1992  8800
13           Central_Forests 1992 20500
14           Western_Forests 1992  6800
15 EasternHyperOceanicBarrens 1992  4400
16          Maritime_Barrens 1992 21300
17            Avalon_Forests 1992  6400
```

19	North_Shore_Forests	1980	8400
20	Northern_Peninsula_Forests	1980	12000
21	Long_Range_Barrens	1980	8800
22	Central_Forests	1980	20500
23	Western_Forests	1980	6800
24	EasternHyperOceanicBarrens	1980	4400
25	Maritime_Barrens	1980	21300
26	Avalon_Forests	1980	6400
28	North_Shore_Forests	1960	8400
29	Northern_Peninsula_Forests	1960	12000
30	Long_Range_Barrens	1960	8800
31	Central_Forests	1960	20500
32	Western_Forests	1960	6800
34	Maritime_Barrens	1960	21300
35	Avalon_Forests	1960	4200
37	North_Shore_Forests	1940	8400
38	Northern_Peninsula_Forests	1940	12000
40	Central_Forests	1940	20500
41	Western_Forests	1940	6800
50	Western_Forests	1904	6800

	Estimated_Moose_Pop	Observation_Method
1	21740	Aerial_Survey
2	32000	Aerial_Survey
3	5300	Camera_Trap
4	41250	Aerial_Survey
5	17000	Expert_Estimate
6	2200	Expert_Estimate
7	20450	Expert_Estimate
8	4040	Aerial_Survey
9	5	Camera_Trap
10	14000	Aerial_Survey
11	21000	Aerial_Survey
12	1540	Expert_Estimate
13	27600	Aerial_Survey
14	11650	Aerial_Survey
15	1400	Camera_Trap
16	14500	Aerial_Survey
17	5320	Expert_Estimate
19	7400	Aerial_Survey
20	15700	Expert_Estimate
21	400	Camera_Trap
22	13200	Aerial_Survey
23	3500	Aerial_Survey
24	600	Expert_Estimate

25	8300	Hunter_Reports
26	200	Camera_Trap
28	10600	Aerial_Survey
29	18000	Aerial_Survey
30	1300	Hunter_Reports
31	17210	Aerial_Survey
32	14500	Expert_Estimate
34	10300	Aerial_Survey
35	200	Expert_Estimate
37	3500	Expert_Estimate
38	9600	Expert_Estimate
40	11200	Expert_Estimate
41	2300	Expert_Estimate
50	4	Count

	Data_Source
1	GovOfNewfoundland
2	NGO_Monitoring
3	MemorialUniversity
4	GovOfNewfoundland
5	ParksCanada
6	MemorialUniversity
7	MemorialUniversity
8	MemorialUniversity
9	ParksCanada
10	GovOfNewfoundland
11	NGO_Monitoring
12	MemorialUniversity
13	GovOfNewfoundland
14	GovOfNewfoundland
15	ParksCanada
16	GovOfNewfoundland
17	MemorialUniversity
19	GovOfNewfoundland
20	MemorialUniversity
21	ParksCanada
22	GovOfNewfoundland
23	GovOfNewfoundland
24	MemorialUniversity
25	GovOfNewfoundland
26	ParksCanada
28	GovOfNewfoundland
29	GovOfNewfoundland
30	GovOfNewfoundland
31	GovOfNewfoundland

```
32       ParksCanada
34  GovOfNewfoundland
35       ParksCanada
37  GovOfNewfoundland
38  GovOfNewfoundland
40       ParksCanada
41       ParksCanada
50  GovOfNewfoundland
> moose_clean <- na.omit(moosedata)
> #
> View(moose_clean)
> #
> # QUESTION 4 ==========================
> select(moose_clean, Ecoregion, Year, Area, Estimated_Moose_Pop)
1        North_Shore_Forests 2020  8400
2  Northern_Peninsula_Forests 2020 12000
3         Long_Range_Barrens 2020  8800
4            Central_Forests 2020 20500
5            Western_Forests 2020  6800
6  EasternHyperOceanicBarrens 2020  4400
7           Maritime_Barrens 2020 21300
8             Avalon_Forests 2020  4200
9   StraitOfBelleIsleBarrens 2020   500
10       North_Shore_Forests 1992  8400
11 Northern_Peninsula_Forests 1992 12000
12        Long_Range_Barrens 1992  8800
13           Central_Forests 1992 20500
14           Western_Forests 1992  6800
15 EasternHyperOceanicBarrens 1992  4400
16          Maritime_Barrens 1992 21300
17            Avalon_Forests 1992  6400
19       North_Shore_Forests 1980  8400
20 Northern_Peninsula_Forests 1980 12000
21        Long_Range_Barrens 1980  8800
22           Central_Forests 1980 20500
23           Western_Forests 1980  6800
24 EasternHyperOceanicBarrens 1980  4400
25          Maritime_Barrens 1980 21300
26            Avalon_Forests 1980  6400
28       North_Shore_Forests 1960  8400
29 Northern_Peninsula_Forests 1960 12000
30        Long_Range_Barrens 1960  8800
31           Central_Forests 1960 20500
32           Western_Forests 1960  6800
```

34	Maritime_Barrens	1960	21300
35	Avalon_Forests	1960	4200
37	North_Shore_Forests	1940	8400
38	Northern_Peninsula_Forests	1940	12000
40	Central_Forests	1940	20500
41	Western_Forests	1940	6800
50	Western_Forests	1904	6800

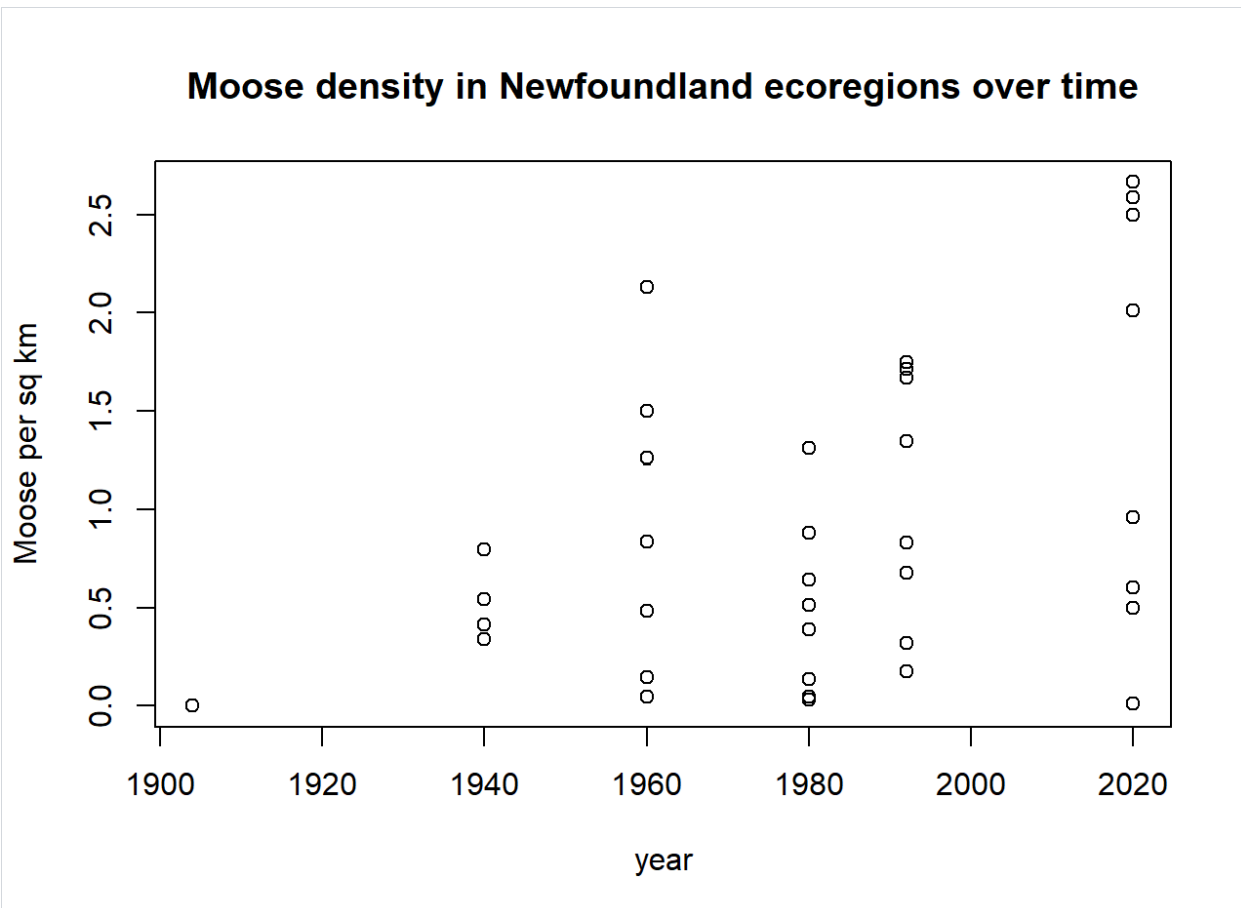
	Estimated_Moose_Pop
1	21740
2	32000
3	5300
4	41250
5	17000
6	2200
7	20450
8	4040
9	5
10	14000
11	21000
12	1540
13	27600
14	11650
15	1400
16	14500
17	5320
19	7400
20	15700
21	400
22	13200
23	3500
24	600
25	8300
26	200
28	10600
29	18000
30	1300
31	17210
32	14500
34	10300
35	200
37	3500
38	9600
40	11200
41	2300

```
50                  4
> moose_sel <- select(moose_clean, Ecoregion, Year, Area, Estimated_Moose_Pop)
> #
> View(moose_sel)
> #
> # QUESTION 5 (a&b) ====================
> # Observations in dataset -------------
> min(moose_sel$Year)
[1] 1904
> year_min <- min(moose_sel$Year)
> #
> max(moose_sel$Estimated_Moose_Pop)
[1] 41250
> moose_max <- max(moose_sel$Estimated_Moose_Pop)
> #
> # QUESTION 6 =========================
> # Calculating moose density -----------
> mutate(moose_sel, MooseDensity = Estimated_Moose_Pop / Area)
> moosedata2 <- mutate(moose_sel, MooseDensity = Estimated_Moose_Pop / Area)
                 Ecoregion Year  Area
1        North_Shore_Forests 2020  8400
2  Northern_Peninsula_Forests 2020 12000
3         Long_Range_Barrens 2020  8800
4            Central_Forests 2020 20500
5            Western_Forests 2020  6800
6  EasternHyperOceanicBarrens 2020  4400
7           Maritime_Barrens 2020 21300
8            Avalon_Forests 2020  4200
9   StraitOfBelleIsleBarrens 2020   500
10       North_Shore_Forests 1992  8400
11 Northern_Peninsula_Forests 1992 12000
12        Long_Range_Barrens 1992  8800
13           Central_Forests 1992 20500
14           Western_Forests 1992  6800
15 EasternHyperOceanicBarrens 1992  4400
16          Maritime_Barrens 1992 21300
17            Avalon_Forests 1992  6400
19       North_Shore_Forests 1980  8400
20 Northern_Peninsula_Forests 1980 12000
21        Long_Range_Barrens 1980  8800
22           Central_Forests 1980 20500
23           Western_Forests 1980  6800
24 EasternHyperOceanicBarrens 1980  4400
25          Maritime_Barrens 1980 21300
```

26	Avalon_Forests	1980	6400
28	North_Shore_Forests	1960	8400
29	Northern_Peninsula_Forests	1960	12000
30	Long_Range_Barrens	1960	8800
31	Central_Forests	1960	20500
32	Western_Forests	1960	6800
34	Maritime_Barrens	1960	21300
35	Avalon_Forests	1960	4200
37	North_Shore_Forests	1940	8400
38	Northern_Peninsula_Forests	1940	12000
40	Central_Forests	1940	20500
41	Western_Forests	1940	6800
50	Western_Forests	1904	6800

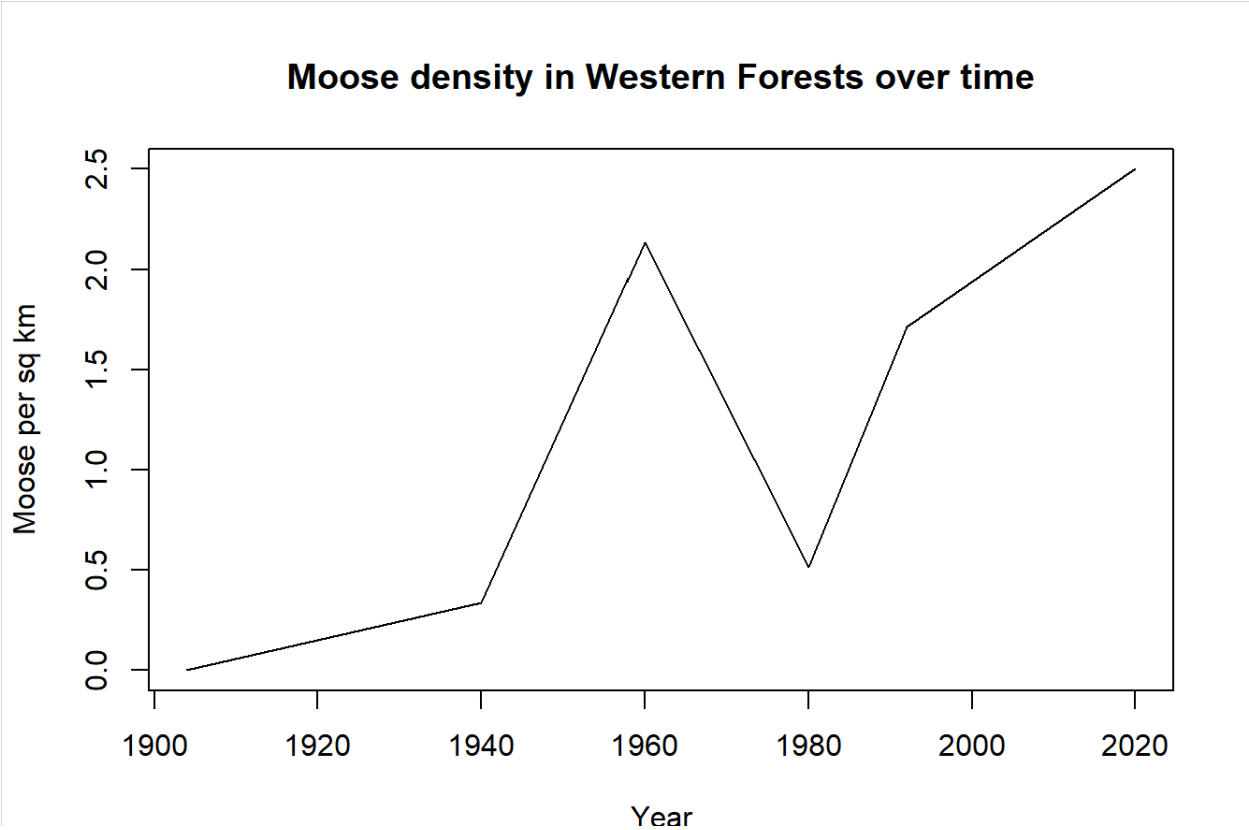
	Estimated_Moose_Pop	MooseDensity
1	21740	2.5880952381
2	32000	2.6666666667
3	5300	0.6022727273
4	41250	2.0121951220
5	17000	2.5000000000
6	2200	0.5000000000
7	20450	0.9600938967
8	4040	0.9619047619
9	5	0.0100000000
10	14000	1.6666666667
11	21000	1.7500000000
12	1540	0.1750000000
13	27600	1.3463414634
14	11650	1.7132352941
15	1400	0.3181818182
16	14500	0.6807511737
17	5320	0.8312500000
19	7400	0.8809523810
20	15700	1.3083333333
21	400	0.0454545455
22	13200	0.6439024390
23	3500	0.5147058824
24	600	0.1363636364
25	8300	0.3896713615
26	200	0.0312500000
28	10600	1.2619047619
29	18000	1.5000000000
30	1300	0.1477272727
31	17210	0.8395121951
32	14500	2.1323529412

```
34            10300 0.4835680751
35              200 0.0476190476
37             3500 0.4166666667
38             9600 0.8000000000
40            11200 0.5463414634
41             2300 0.3382352941
50                4 0.0005882353
> #
> View(moosedata2)
>#
> # QUESTION 7 ==========================
> plot(moosedata2$Year, moosedata2$MooseDensity,
  +     xlab = "year",
  +     ylab = "Moose per sq km",
  +     main = "Moose density in Newfoundland ecoregions over time")
> #
```




```
> # QUESTION 8 ==========================
> # Western forests data-----------------
```

```
> filter(moosedata2, Ecoregion == "Western_Forests")
> moose_west <- filter(moosedata2, Ecoregion == "Western_Forests")
> #
> # Showing change in moose density------
> plot(moose_west$Year, moose_west$MooseDensity,
    +     type = "l",
    +     xlab = "Year",
    +     ylab = "Moose per sq km",
    +     main = "Moose density in Western Forests over time")
```



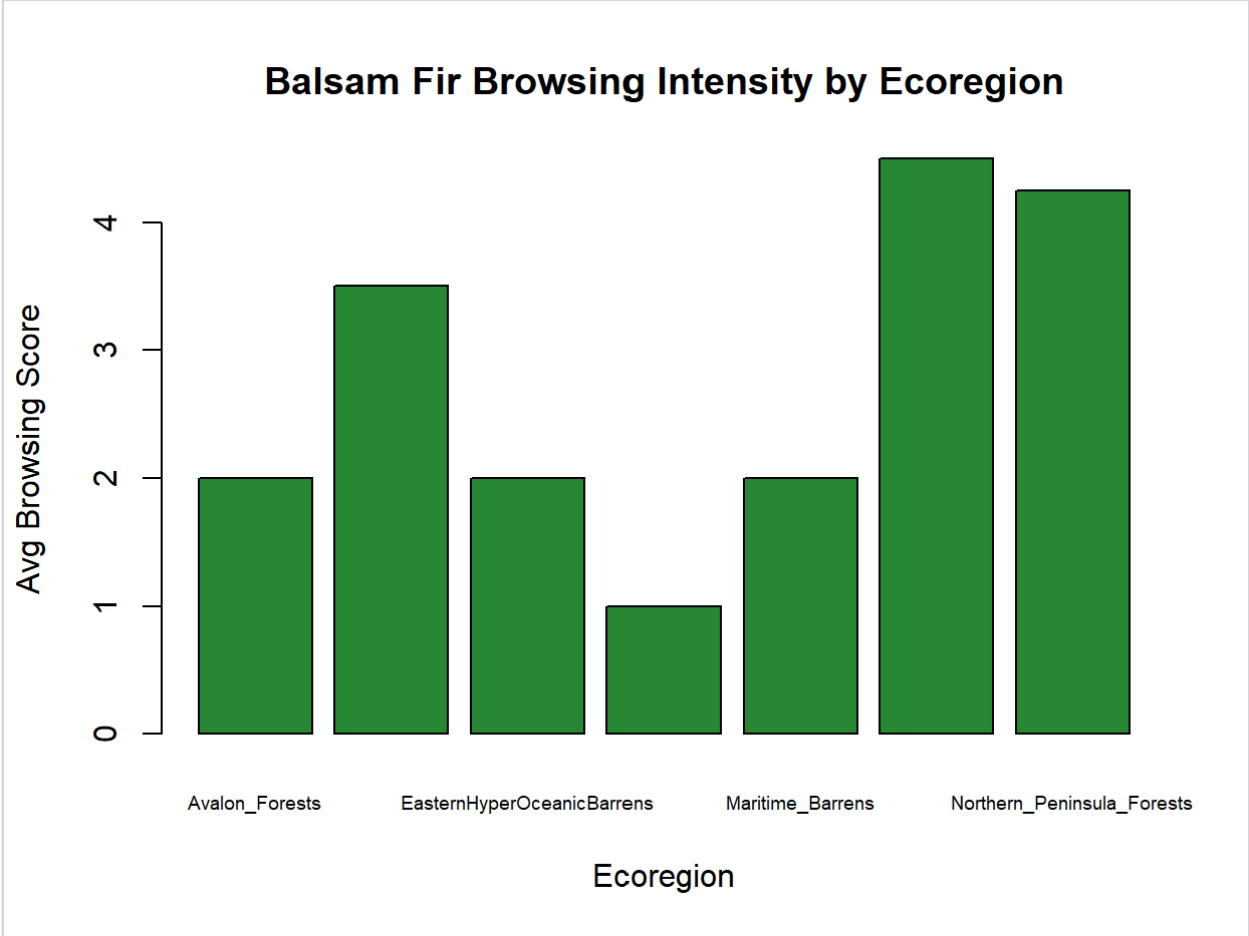

```
> #
> # QUESTION 9 ==========================
> # Saving 2020 dataset -----------------
> filter(moosedata2, Year == 2020)
> moose_2020 <- filter(moosedata2, Year == 2020)
> #
> # High densities in 2020 dataset-------
> filter(moose_2020, MooseDensity > 2.0)
> moose_2020_high <- filter(moose_2020, MooseDensity > 2.0)
#
> # Arranging data ----------------------
> arrange(moose_2020_high, desc(MooseDensity))
> moose_2020_high_byD <- arrange(moose_2020_high, desc(MooseDensity))
```

```
> #
> # QUESTION 10 ==========================
> # Using pipes ------------------------
> #
  > moosedata2 %>%
  +     filter(Year == 2020) %>%
  +     filter(MooseDensity > 2.0) %>%
  +     arrange(desc(MooseDensity)) %>%
  +     print()
> moosefinal <- moosedata2 %>%
  +     filter(Year == 2020) %>%
  +     filter(MooseDensity > 2.0) %>%
  +     arrange(desc(MooseDensity)) %>%
  +     print()
> #
> # QUESTION 11 ==========================
> # Naming Data file --------------------
> saplings <- read.csv("C:/Users/KatieTurpin/Downloads/SaplingStudy.csv")
> #
> # Omitting data ------------------------
> View(saplings)
> na.omit(saplings)
> sap_clean <- na.omit(saplings)
> #
> # QUESTION 12 ==========================
> # Moose browsing variation -------------
> sap_clean %>%
  +     group_by(Ecoregion) %>%
  +     summarize(AverageBrowsing = mean(BrowsingScore)) %>%
sap_reg_browse <- sap_clean %>%
  +     group_by(Ecoregion) %>%
  +     summarize(AverageBrowsing = mean(BrowsingScore)) %>%
  +     print()
# A tibble: 9 × 2
Ecoregion                 AverageBrowsing
<chr>                           <dbl>
1 Avalon_Forests                     2
2 Central_Forests                    4
3 EasternHyperOceanicBarrens           2.4
4 Long_Range_Barrens                   2.6
5 Maritime_Barrens                   1.83
6 North_Shore_Forests                  4.38
7 Northern_Peninsula_Forests           4.57
8 StraitOfBelleIsleBarrens             1
```

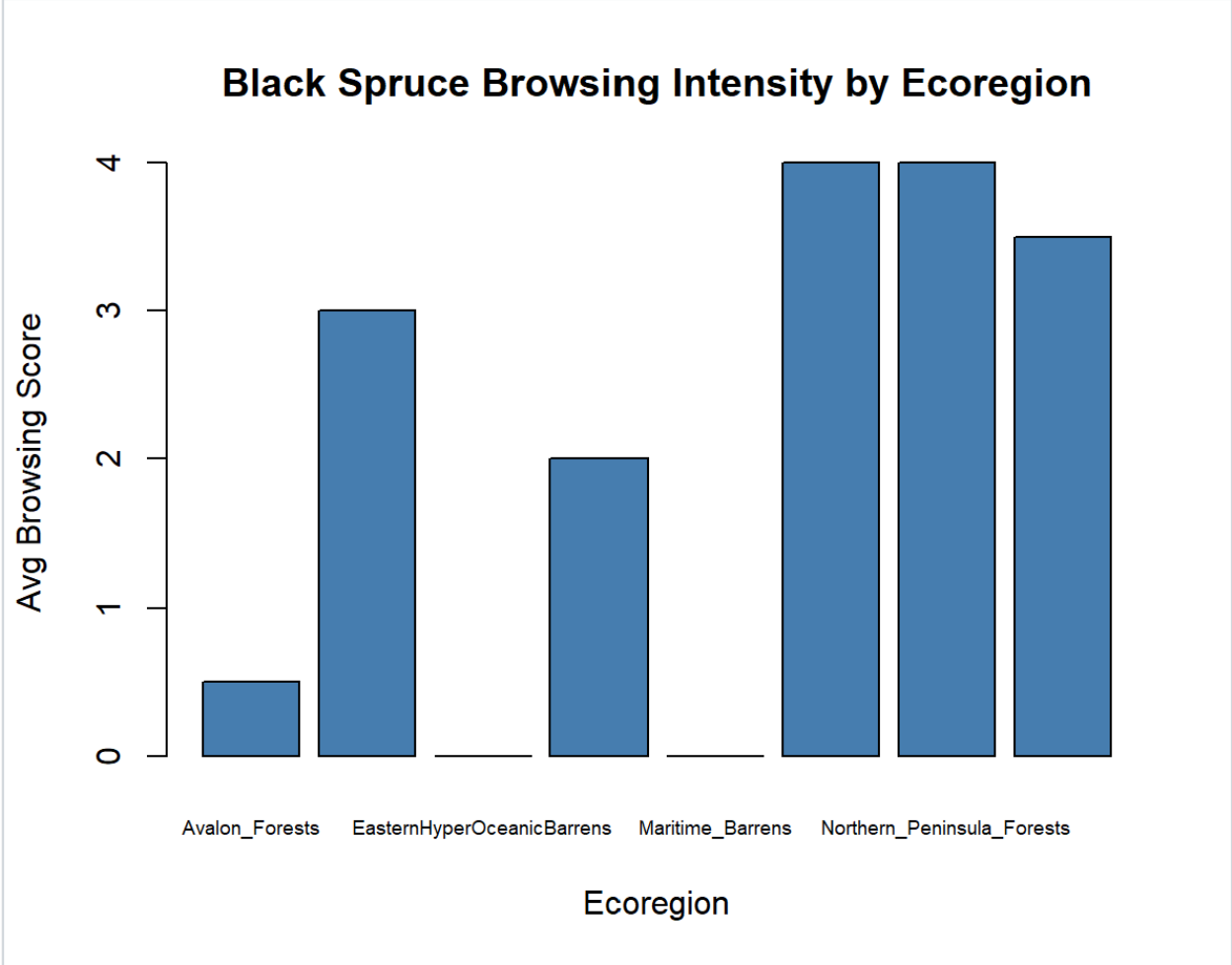
```
9 Western_Forests                    4.5
> #
> # Hghest/Lowest browsing----------------
> sap_reg_browse %>%
  +     arrange(AverageBrowsing)
> sap_reg_browse %>%
  +     arrange(-(AverageBrowsing))
              # A tibble: 9 × 2
  Ecoregion                 AverageBrowsing
  <chr>                               <dbl>
1 Northern_Peninsula_Forests           4.57
2 Western_Forests                     4.5
3 North_Shore_Forests                   4.38
4 Central_Forests                     4
5 Long_Range_Barrens                    2.6
6 EasternHyperOceanicBarrens            2.4
7 Avalon_Forests                      2
8 Maritime_Barrens                     1.83
9 StraitOfBelleIsleBarrens              1
> avg_browse_reg <- sap_reg_browse %>%
  +     arrange(-(AverageBrowsing)
> View(avg_browse_reg)
> # Northern_Peninsula_Forests:Highest, StraitOfBelleIsleBarrens:Lowest
> #
> # QUESTION 13 ==========================
> Tree height variation-------------------
> sap_clean %>%
  +     group_by(Ecoregion) %>%
  +     summarize(AverageHeight = mean(Height)) %>%
  +     print()
# A tibble: 9 × 2
Ecoregion                 AverageHeight
<chr>                             <dbl>
1 Avalon_Forests                    32.4
2 Central_Forests                   23.8
3 EasternHyperOceanicBarrens         31.6
4 Long_Range_Barrens                  29.9
5 Maritime_Barrens                   26.7
6 North_Shore_Forests                 22.3
7 Northern_Peninsula_Forests          19.9
8 StraitOfBelleIsleBarrens           25.4
9 Western_Forests                   18.9
> sap_reg_height <- sap_clean %>%
  +   group_by(Ecoregion) %>%
```

```
  +   summarize(AverageHeight = mean(Height)) %>%
> #
> # Lowest average heights----------------
> sap_reg_height %>%
  +     filter(AverageHeight < 20) %>%
  +     print()
# A tibble: 2 × 2
Ecoregion                  AverageHeight
<chr>                              <dbl>
1 Northern_Peninsula_Forests          19.9
2 Western_Forests                     18.9
> sap_reg_height %>%
  +     filter(AverageHeight < 20) %>%
  +     print()
> # Northern_Peninsula_Forests and Western_Forests are < 20 cm
> #
> # QUESTION 14 ==========================
> sap_clean %>%
  +     group_by(Species) %>%
  +     summarize(AverageBrowsing = mean(BrowsingScore)) %>%
  +     print()
# A tibble: 6 × 2
Species        AverageBrowsing
<chr>                    <dbl>
1 "Alder "                 4.25
2 "Balsam_Fir"             3.14
3 "Black_Ash"              5
4 "Black_Spruce"           2.33
5 "White_Birch"            3.14
6 "Willow"                 4.31
> sap_spe_browse <- sap_clean %>%
  +     group_by(Species) %>%
  +     summarize(AverageBrowsing = mean(BrowsingScore)) %>%
  +     print()
> #
> # Highest/Lowest Browsing --------------
> arrange(sap_spe_browse, desc(AverageBrowsing))
# A tibble: 6 × 2
Species        AverageBrowsing
<chr>                    <dbl>
1 "Black_Ash"              5
2 "Willow"                 4.31
3 "Alder "                 4.25
4 "White_Birch"            3.14
```

```
5 "Balsam_Fir"              3.14
6 "Black_Spruce"            2.33
> avg_browse_spe <- arrange(sap_spe_browse, desc(AverageBrowsing))
> #
> # Black_ash:Highest Black_spruce:Lowest
> #
> # QUESTION 15 ==========================
> # Average browsing variation------------
> sap_clean %>%
  +     filter(Species == "Balsam_Fir") %>%
  +     group_by(Ecoregion) %>%
  +     summarize(AverageBrowsing = mean(BrowsingScore))
# A tibble: 7 × 2
Ecoregion                  AverageBrowsing
<chr>                                <dbl>
1 Avalon_Forests                       2
2 Central_Forests                      3.5
3 EasternHyperOceanicBarrens           2
4 Long_Range_Barrens                   1
5 Maritime_Barrens                     2
6 North_Shore_Forests                  4.5
7 Northern_Peninsula_Forests           4.25
> fir_reg_browse <- sap_clean %>%
  +     filter(Species == "Balsam_Fir") %>%
  +     group_by(Ecoregion) %>%
  +     summarize(AverageBrowsing = mean(BrowsingScore))
> #
> # QUESTION 16 ==========================
> # Making a barplot---------------------
> barplot(fir_reg_browse$AverageBrowsing,
          +         names.arg = fir_reg_browse$Ecoregion,
          +         xlab = "Ecoregion",
          +         ylab = "Avg Browsing Score",
          +         main = "Balsam Fir Browsing Intensity by Ecoregion",
          +         col = "forestgreen",
          +         cex.names = 0.6)
> #
```




```
> # QUESTION 17 ==========================
> # Spruce browsing ---------------------
> spruce_reg_browse <- sap_clean %>%
  +     filter(Species == "Black_Spruce") %>%
  +     group_by(Ecoregion) %>%
  +     summarize(AverageBrowsing = mean(BrowsingScore))
> #
> # Spruce barplot------------------------
> barplot(spruce_reg_browse$AverageBrowsing,
          +         names.arg = spruce_reg_browse$Ecoregion,
          +         xlab = "Ecoregion",
          +         ylab = "Avg Browsing Score",
          +         main = "Black Spruce Browsing Intensity by Ecoregion",
          +         col = "steelblue",
          +         cex.names = 0.6)
```

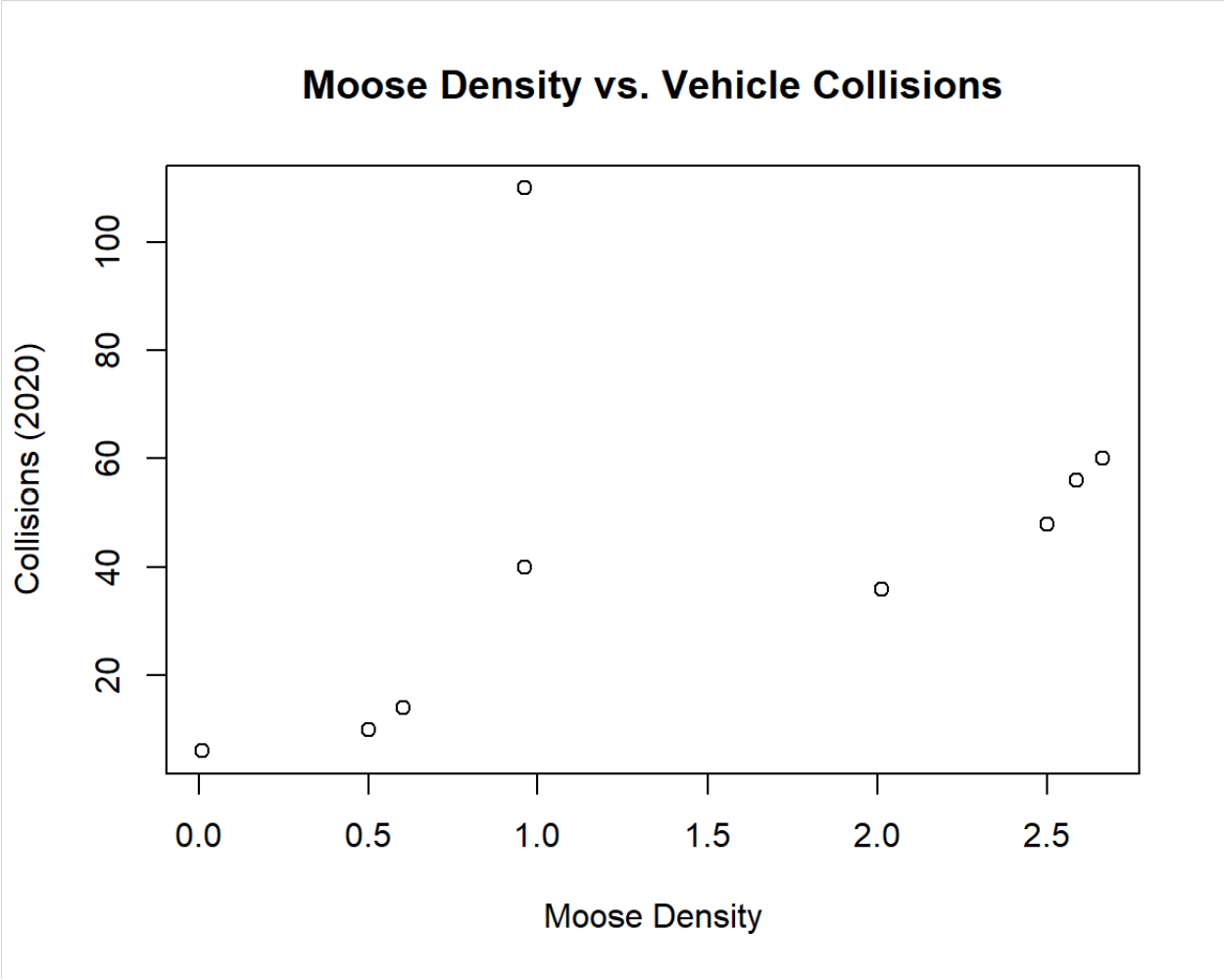



```
> # Comparison----------------------------
> # Balsam fir had overall higher browsing rates. However Black Spruce had higher browsing
rates in Long_Range_Barrens.
> #
> # QUESTION 18 ==========================
> # Tally -------------------------------
> sap_reg_tally<- sap_clean %>%
  +     group_by(Ecoregion) %>%
  +     tally() %>%
  +     print()
# A tibble: 9 × 2
Ecoregion                    n
<chr>                    <int>
1 Avalon_Forests               5
2 Central_Forests              5
3 EasternHyperOceanicBarrens     5
4 Long_Range_Barrens             5
5 Maritime_Barrens               3
```

```
6 North_Shore_Forests            8
7 Northern_Peninsula_Forests     7
8 StraitOfBelleIsleBarrens       1
9 Western_Forests                5
> #
> # QUESTION 19 ==========================
> sap_spe_tally <- sap_clean %>%
  +     group_by(Species) %>%
  +     tally() %>%
  +     print()
# A tibble: 6 × 2
Species            n
<chr>          <int>
1 "Alder "           8
2 "Balsam_Fir"      11
3 "Black_Ash"        1
4 "Black_Spruce"     9
5 "White_Birch"      7
6 "Willow"           8
> #
> # QUESTION 20 ==========================
> # distribution -------------------------
> # No they were not as the two had different numbers of individuals sampled.
> # It is important to recognize bias in datasets because even a small bias can impact averages
and information gathered greatly.
> #
> # QUESTION 21 ==========================
moose_sap <- left_join(moose_2020, sap_clean, by = 'Ecoregion', relationship =
"many-to-many")
> moose_2020b <- moose_clean %>%
  +     filter(Year == 2020) %>%
  +     mutate(MooseDensity = Estimated_Moose_Pop / Area)
> moose_sap <- left_join(moose_2020b, sap_clean, by = 'Ecoregion', relationship =
"many-to-many")
> #
> # QUESTION 22 ==========================
> sum_spe_browse <- moose_sap %>%
  +     group_by(Species, Ecoregion) %>%
  +     summarize(AvgBrowsing = mean(BrowsingScore),
                +               AvgDensity = mean(MooseDensity)) %>%
  +     print()
`summarise()` has regrouped the output.
i Summaries were computed grouped by Species
and Ecoregion.
```

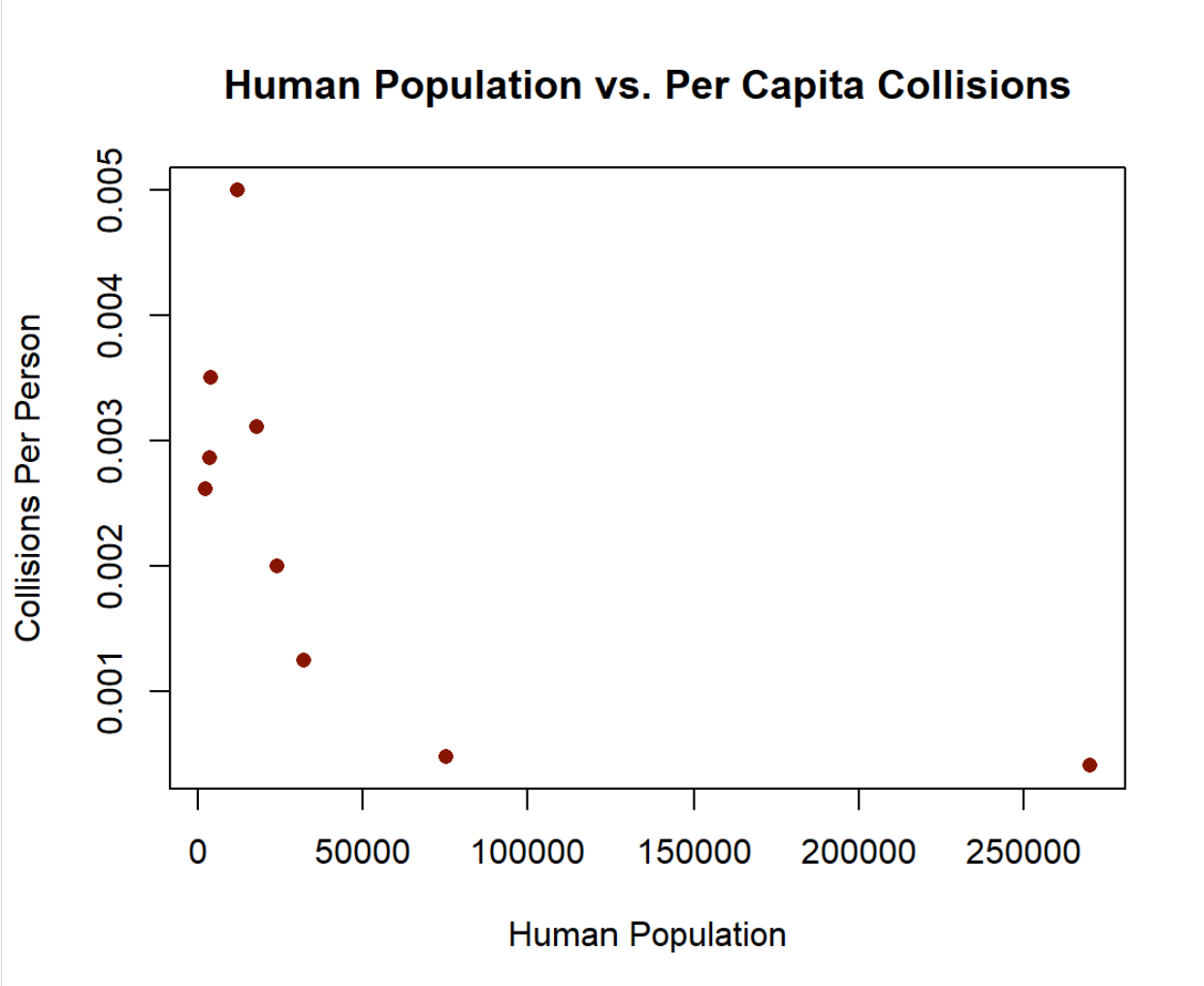
```
i Output is grouped by Species.
i Use `summarise(.groups = "drop_last")` to
silence this message.
i Use `summarise(.by = c(Species, Ecoregion))`
for per-operation grouping instead.
# A tibble: 38 × 4
# Groups:   Species [6]
Species      Ecoregion AvgBrowsing AvgDensity
<chr>        <chr>           <dbl>      <dbl>
1 "Alder "     Avalon_F…         4        0.962
2 "Alder "     Central_…         4.5      2.01
3 "Alder "     EasternH…         4        0.5
4 "Alder "     Long_Ran…         3.5      0.602
5 "Alder "     Maritime…         3.5      0.960
6 "Alder "     North_Sh…         4.5      2.59
7 "Alder "     Northern…         5        2.67
8 "Alder "     Western_…         5        2.5
9 "Balsam_Fir" Avalon_F…         2        0.962
10 "Balsam_Fir" Central_…         3.5      2.01
# i 28 more rows
# i Use `print(n = ...)` to see more rows
> #
> # QUESTION 23 ==========================
> #Hypothesis Evidence: Yes, there is evidence. At lower densities, browsing scores vary wildly
between species (selective), while at higher densities, the scores for all species tend to cluster
higher and more closely together.
> #Preferences: Balsam Fir typically shows the highest browsing scores (most favored), while
Black Spruce often shows the lowest (least favored).
> #Missing Species: Black Ash is often missing from these figures because it had a sample size
of only 1, making it impossible to calculate a meaningful average for density correlation.
> #
> # QUESTION 24 ==========================
> collisions2020 <- c(56, 60, 14, 36, 48, 10, 40, 110, 6)
> human_pop <- c(18000, 12000, 4000, 75100, 24000, 3500, 32000, 270000, 2300)
> study_sites <- c("North_Shore_Forests","Northern_Peninsula_Forests",
"Long_Range_Barrens","Central_Forests","Western_Forests","EasternHyperOceanicBarrens","
Maritime_Barrens","Avalon_Forests","StraitOfBelleIsleBarrens")
moose_coll <- data.frame(collisions2020, human_pop, study_sites)
> #

> # QUESTION 25 ==========================
> moose_coll2 <- moose_coll %>%
  +     rename(Ecoregion = study_sites)
> coll_merge <- left_join(moose_coll2, moose_2020, by = "Ecoregion")
```

```
> #
> # QUESTION 26 ==========================
> plot(coll_merge$MooseDensity, coll_merge$collisions2020,
+     xlab = "Moose Density", ylab = "Collisions (2020)",
+     main = "Moose Density vs. Vehicle Collisions")
```




```
> # QUESTION 26 ==========================
> plot(coll_merge$MooseDensity, coll_merge$collisions2020,
+     xlab = "Moose Density", ylab = "Collisions (2020)",
+     main = "Moose Density vs. Vehicle Collisions")
> # Yes there is an outlier where the moose density is 1 and the # of collisions is over 100.
Other than that as moose density increases collisions increase.
> #
> # QUESTION 27 ==========================
> coll_merge_per_capita <- coll_merge %>%
  +     mutate(coll_per_capita = collisions2020 / human_pop)
```

```
> #
> # QUESTION 28 ==========================
> plot(coll_merge_per_capita$human_pop, coll_merge_per_capita$coll_per_capita,
+     xlab = "Human Population",
+     ylab = "Collisions Per Person",
+     main = "Human Population vs. Per Capita Collisions",
+     pch = 16,
+     col = "darkred")
> #
```




```
> # QUESTION 29 ==========================
> # As the human population increases, collisions per person decreases. This makes sense because as populations increase we move towards more urbanized areas with less wildlife.
```